Choose your robot’s next job

Name: ____________________ Team: ____________________ Date: __________

Choose ONE build. Reuse the wheeled base you have already tested. Your teacher must approve its attachment points and each new setup before you run it.

Choice	What you build	Start in this packet
1. Parcel delivery robot	Carry a paper parcel to a red-line delivery space.	Pages 3-4
2. Drawing robot	Draw a short line on a protected paper surface.	Pages 5-6
3. Paper sweeper	Push three light paper pieces into a collection area.	Pages 7-8
4. Ball-pushing robot	Guide one light paper ball toward a goal.	Pages 9-10
5. Rescue carrier	Tow a light paper figure on a small rescue sled.	Pages 11-12

Our choice: ____

Why we chose it: ____

Before you build

[ ] Our shared base passed the short-drive and Stop checks.

[ ] We measured the clear mounting space.

[ ] We found fixed attachment points with the teacher.

[ ] We have the materials for one choice, not all five.

Mounting points means the places where you attach a new part. Use fixed parts of the frame. Do not fasten to wheels, axles, sensors, buttons or moving motor parts.

If the card sizes do not fit, make them smaller. Ask the teacher to approve the changes. These attachment plans are original teaching drafts and still need a real-kit tryout.

Plan, test and share

Name: ____________________ Team: ____________________ Date: __________

First class: choose and build

[ ] Read your choice's two pages. Explain the robot's new job.

[ ] Gather the materials. Build one step at a time with the power off.

[ ] Check the new attachment with your teacher. Try one short run.

Second class: test and improve

[ ] Use your choice's test table for three runs. Write every result.

[ ] Change one thing and try again. Keep the original notes.

[ ] Save the final code with a clear name.

[ ] Ask another team to follow your setup, start and stop directions.

Shared check before every run

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Show what you learned

Give a two-minute presentation: explain the job, show how the program works, compare results before and after a change, then name one next test.

My job on the team: ____

One choice I made: ____

A result that helped me decide: _______________________________

A run passes only if it meets every rule for your chosen build. Aim for at least two passing runs out of three. All runs must keep the Stop controls available.

Choice 1: Parcel delivery robot

Name: ____________________ Team: ____________________ Date: __________

Carry a paper parcel to a red-line delivery space.

Gather these materials

[ ] The shared checked base

[ ] Thin card about 12 by 8 cm

[ ] One paper parcel about 4 by 3 by 2 cm

[ ] Ruler, scissors and removable tape

Tray plan: cut away the four corner squares.

Fold up the four sides on the dashed lines.

Start with a 1 cm wall. Resize the base to fit.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

[ ] 1. Measure the clear mounting space on the base. Cut the card to fit it. Keep at least 1 cm of room between the attachment and any wheel. Make it smaller if needed.

[ ] 2. Draw a fold line 1 cm inside each edge. Cut out the four corner squares. Fold all four walls up and tape the corners to make a shallow tray.

[ ] 3. Fold a light paper parcel. Place it in the center of the tray. Add a small folded-paper stop at each end if it slides. Do not add heavy contents.

[ ] 4. Turn the robot off. Use the teacher-approved mounting points and removable connectors to hold the tray flat. Keep buttons and the sensor clear.

[ ] 5. Raise the wheels and check for rubbing. Then place the robot on the white lane from lesson 6. Keep the broad red strip and empty space beyond it.

[ ] 6. Run the code below three times. If the parcel falls, change one tray feature and repeat. Unload the parcel by hand only after the robot has stopped.

Choice 1: program and tests

Name: ____________________ Team: ____________________ Date: __________

Make the program

1. Use lesson 6’s tested color-stop program without removing its limits.

2. Start stopped; set keepGoing to Yes. Repeat at most 20 times. Only while keepGoing is Yes, read the color: white means a slow 0.1-second move then Stop; any other reading means Stop and set keepGoing to No.

3. Stop and end after the repeats. Red means delivery; unknown means check the setup.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

[ ] The parcel stays in the tray.

[ ] The whole robot stops inside the delivery space.

[ ] The robot hits no object or person.

Keep the same setup for three runs

Run	Parcel stayed on?	Inside space?	Hit nothing?
1			
2			
3			

One change to try

Which tray feature helped the parcel stay on?

My change: ____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Choice 2: Drawing robot

Name: ____________________ Team: ____________________ Date: __________

Draw a short line on a protected paper surface.

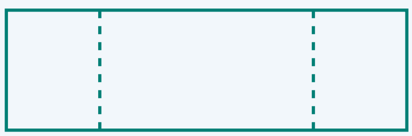
Gather these materials

[ ] The shared checked base

[ ] One washable felt-tip marker

[ ] A card strip about 12 by 3 cm and removable tape

[ ] A large sheet of paper over a protective mat



Wrap this card around the marker.

Tape card to card to make a sleeve.

The marker tip should just touch the paper.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

[ ] 1. Cover a large floor area with a protective mat and paper. Tape the paper edges flat. Ask the teacher to approve the marker and surface.

[ ] 2. Wrap the card strip loosely around the marker to make a sleeve. Tape the card to itself. The marker must slide up and down inside it.

[ ] 3. Turn the robot off. Fix the sleeve upright at a teacher-approved point behind the wheels. Keep it away from the sensor and moving parts.

[ ] 4. Slide the uncapped marker down until its tip just touches the paper. Secure the marker in the sleeve with a small removable tape tab. It must not lift the wheels off the paper.

[ ] 5. Raise the wheels and check the short program. Put the robot back on the protected paper with enough space for the whole run.

[ ] 6. Draw three short lines. Stop and turn off the robot before lifting or adjusting the marker. If a line has gaps, lower the tip a little; if the robot struggles, raise it.

Choice 2: program and tests

Name: ____________________ Team: ____________________ Date: __________

Make the program

1. Start with movement stopped. Use a checked low speed, such as 20 percent.

2. Drive forward for 0.8 seconds, then stop and end. Shorten the time if the paper is too small.

3. No automatic repeat. Lift the marker and return the robot by hand between runs with the power off.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

[ ] A visible line is drawn on the paper.

[ ] The marker stays in its holder and all marks stay on protected paper.

[ ] The robot stops by itself at the programmed time.

Keep the same setup for three runs

Run	Visible line?	Stayed on paper?	Stopped by itself?
1			
2			
3			

One change to try

How did marker height change the line or the robot’s movement?

My change: ____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Choice 3: Paper sweeper

Name: ____________________ Team: ____________________ Date: __________

Push three light paper pieces into a collection area.

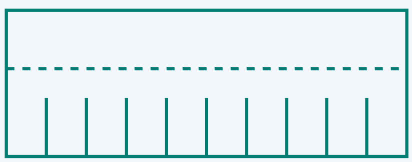
Gather these materials

[ ] The shared checked base

[ ] A thin card strip about 12 by 5 cm

[ ] Three tissue-paper squares, each about 2 cm wide

[ ] Removable tape, ruler and scissors



Fold the top strip back for mounting.

Cut slits from the bottom to make bristles.

Keep every slit below the fold line.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

[ ] 1. Cut a card strip as wide as the robot, but no wider than the clear test lane. Fold its top 2 cm backward to make a mounting edge.

[ ] 2. Cut short vertical slits about 1 cm apart into the lower 2 cm. These make flexible paper bristles. Round sharp card corners with scissors.

[ ] 3. Turn the robot off. Attach the folded top edge to approved fixed front mounting points. The flexible ends should just brush the floor without lifting any wheel.

[ ] 4. Use a clear floor lane. Mark a collection box farther along it. Place three tissue squares in a row across the starting sweep area. Keep them away from wheel paths.

[ ] 5. Try the short program below. Check that the paper bristles can bend. Stop at once if a piece catches in a wheel; turn the power off before removing it.

[ ] 6. Count the pieces that reach the box. Repeat three times with the same starting positions. Adjust the bristle height once and compare.

Choice 3: program and tests

Name: ____________________ Team: ____________________ Date: __________

Make the program

1. Start stopped and set the checked low speed.

2. Drive forward for 0.8 seconds, then stop and end. Set the collection box using the distance measured in lesson 2.

3. Do not add continuous driving. Return the paper pieces and robot to their marks between tests.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

[ ] At least two of the three tissue squares reach the box.

[ ] No paper gets caught in the wheels.

[ ] The robot stops by itself and touches no object except the tissue squares.

Keep the same setup for three runs

Run	Pieces in box (0-3)	Wheels clear?	Stopped by itself?
1			
2			
3			

One change to try

Did shorter or longer bristles move the paper better?

My change: ____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Choice 4: Ball-pushing robot

Name: ____________________ Team: ____________________ Date: __________

Guide one light paper ball toward a goal.

Gather these materials

[ ] The shared checked base

[ ] Thin card about 14 by 5 cm

[ ] One crumpled paper ball about 3 cm wide

[ ] Removable tape and ruler

Top view: a shallow V guides the paper ball.

The opening faces forward.

Keep room for the ball to roll freely.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

[ ] 1. Fold the card lengthwise so it has a 3 cm upright wall and a 2 cm mounting flap. Fold it gently across the middle to make a shallow V.

[ ] 2. Make one light paper ball about 3 cm wide. Put it in the V opening. Leave enough room for it to roll freely; do not make a tight clamp.

[ ] 3. Turn the robot off. Attach the two mounting flaps to teacher-approved fixed front points. Keep the V open at the front. Leave the wheels clear and keep the card just above the floor.

[ ] 4. Keep the sensor clear even though this task uses a timed drive. Mark a wide goal across a short straight lane. Place the ball at the same start mark each time.

[ ] 5. Use the short program below. The robot pushes the ball; it does not grip it. Stop if the ball rolls toward people or outside the clear test area.

[ ] 6. Repeat three times. If the ball escapes sideways, adjust the V angle a little. Keep the same ball and code while comparing.

Choice 4: program and tests

Name: ____________________ Team: ____________________ Date: __________

Make the program

1. Start stopped and use the checked low speed.

2. Drive forward for 0.8 seconds, then stop and end. Put the goal within the tested travel distance.

3. No automatic repeat. Return the robot and ball to their marks with the power off.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

[ ] The ball crosses the marked goal line.

[ ] The guide stays attached and the wheels turn freely.

[ ] The robot stops by itself and touches only the paper ball.

Keep the same setup for three runs

Run	Ball crossed goal?	Guide stayed on?	Stopped by itself?
1			
2			
3			

One change to try

Which V angle kept the ball in front of the robot?

My change: ____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Choice 5: Rescue carrier

Name: ____________________ Team: ____________________ Date: __________

Tow a light paper figure on a small rescue sled.

Gather these materials

[ ] The shared checked base

[ ] Thin card: one 12 by 6 cm piece and one 12 by 2 cm strip

[ ] A flat paper person, two paper strips for bands, removable tape and scissors

[ ] Ruler and a clear straight lane

Sled: fold the long sides up.

Wide tow bar: attach at the front.

Keep the bar away from the robot wheels.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

[ ] 1. Use the 12 by 6 cm card as a sled. Fold each long edge up by 1 cm. Curl the front edge slightly upward so it does not catch on the floor.

[ ] 2. Draw and cut out a small paper person. Lay it flat on the sled. Add two loose paper bands across the sled and tape their ends to the sides to hold the figure in place.

[ ] 3. Use the 12 by 2 cm card strip as a wide tow bar. Tape one end flat to the sled’s front. Make a small folded tab at the other end.

[ ] 4. Before attaching it to the robot, pull the sled gently by hand using its wide tow bar. It should slide without catching. Keep it light and use only a straight route for this task.

[ ] 5. Turn the robot off. Ask the teacher to connect the tow-bar tab to a fixed rear mounting point. Keep the whole strip centered behind the robot and away from the wheels. Do not use loose string.

[ ] 6. Run the short program below three times. If the sled catches or a wheel struggles, stop. Smooth the sled edge or shorten the route, then test again.

Choice 5: program and tests

Name: ____________________ Team: ____________________ Date: __________

Make the program

1. Start stopped and use the checked low speed.

2. Drive straight for 0.8 seconds, then stop and end. Do not add turns or reversing for this first sled build.

3. Return the robot and sled to the start with the power off. Do not move a real person, animal or heavy object.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

[ ] The paper person stays on the sled.

[ ] The tow bar stays attached and away from the wheels.

[ ] The robot moves the sled and stops by itself.

Keep the same setup for three runs

Run	Person stayed on?	Tow bar clear?	Moved and stopped?
1			
2			
3			

One change to try

What helped the sled slide without catching?

My change: ____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.